Department of Information and Communication Technology

Faculty of Technology

University of Ruhuna

TecSystem

System for Faculty of Technology / University of Ruhuna

Object Oriented Programming Practicum – ICT 2132

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CONTENT

1. Introduction

1.1 Purpose
1.2 Scope
1.3 Overview

2. Description

2.1 System Objectives
2.2 System Functions
2.3 User Classes and Characteristics
2.4 Operating Environment
2.5 Design and Implementation Constraints

3. Specific Requirement

3.1 Functional Requirement

3.1.1 Admin

3.1.2 Lecturer

3.1.3 Technical Officer

3.1.4 Undergraduates

3.1.5 Non-Functional Requirements

4.Performance Requirement

4.1 Security Requirements

4.2 Usability Requirements

4.3 Compatibility

5. External Interface Requirement

6. Additional Requirement

6.1 Accessibility

6.2 Internationalization

7.References

1.INTRODUCTION

1.1 Purpose

The purpose of this SRS is to establish a clear framework for the development of “TecSystem” a digital management portal. This document details the functional and non-functional requirements to ensure the system effectively manages academic and administrative workflows.

1.2 Scope

Tecsystem is a java-based desktop application integrated with a MYSQL database. It automates user profile management, student attendance tracking, examination marks processing, and eligibility calculations for the Faculty of Technology.

1.3 Overview

The system provides tailored interfaces for four user roles: Admin, Lecturer, Technical Officer and Undergraduate. It replaces manual record-keeping with an automated, secure and role-based digital environment.

2.DESCRIPTION

2.1 System Objectives

- Role-Based Access : To provide secure, specific functionalities based on the user's role.

- Academic Automation : To calculate GPA and exam eligibility automatically based on university criteria.

- Data Centralization : To maintain a single source of truth for attendance, medicals, and marks.

2.2 System Functions

- User Management : Creating and maintaining profiles for staff and students.

- Attendance Tracking : Recording theory and practical sessions for a 15-week semester.

- Result Processing : Calculating CA marks and final grades based on current circulars.

- Communication : Publishing notices and department-specific timetables.

- Medical Records Handling : System maintain a logs medical records for academic and eligibility adjustments.

2.3 User Classes and Characteristics

User class	Key Responsibilities
Admin	Manage Courses, Manage User Accounts, View Grades, Manage & View Attendance, Manage & View Timetables, Approve Medicals, Post & View Notices
Lecturer	Manage Courses, Enter & Update Marks, View Grades, Manage & View Attendance, Post & View Notices, View Timetables
Undergraduate	View Grades, View Attendance, View Notices, View Timetables, Submit Medicals, View Course Details, Update profile Details
Technical Officer	Post & View Notices, Manage & View Timetables, Maintain Attendance Details, Maintain Medical details

2.4 Operating Environment

- Platform : Windows or Linux with Java Runtime Environment (JRE).
- Database : MySQL Server for data storage.
- Hardware : Minimum 8GB RAM and Quad-core CPU for optimal GUI performance.

2.5 Design and Implementation Constraints

- Language : Must be implemented using Java with a focus on OOP principles.(Inheritance, Polymorphism etc)
- Database : Must use MySQL for all persistent storage.
- Policy : Must adhere to the UGC Commission Circular No. 12-2024 for grading.

3.SPECIFIC REQUIREMENT

3.1 Functional Requirement

3.1.1 Admin

- Create, Update and delete all user profiles.
- Add and maintain courses, notices and timetables.

3.1.2 Lecturer

- Upload and modify course materials.
- Input exam marks and generate GPA/SGPA reports.
- View student eligibility based on attendance and CA marks.

3.1.3 Technical Officer

- Daily entry and maintenance of undergraduate attendance.
- Upload and manage medical records, marking them as approved or pending.

3.1.4 Undergraduates

- Update personal contact details and profile pictures.
- Access and view attendance, medical status, grades and timetables.

3.1.5 Non-Functional Requirements

- **Scalability :** Support for at least 20 undergraduates and multiple staff roles per batch.
- **Performance :** Queries for eligibility and marks should execute in under 2 seconds.

4. PERFORMANCE REQUIREMENTS

4.1 Security Requirements

- Encryption of sensitive data like student marks.
- Hashed passwords and secure login protocols to prevent unauthorized access

4.2 Usability Requirements

- Intuitive Java-based GUI with clear tooltips and error messages.
- Users should be able to navigate the core functions within 5 minutes of introduction.

4.3 Compatibility

- Seamless operation across different screen resolutions and OS environments (Windows/ Linux)

5. EXTERNAL INTERFACE REQUIREMENTS

- **User Interface :** A consistent GUI developed using Java Swing or JavaFX.
- **Software Interface** : JDBC (Java Database Connectivity) for MySQL communication.
- **Reporting Interface :** Capability to generate PDF versions of student grade sheets.

6. ADDITIONAL REQUIREMENTS

6.1 Accessibility

- Support for keyboard-only navigation to assist users with motor impairments.
- Descriptive error handling to guide users through incorrect inputs.

6.2 Internationalization

- Support for Unicode characters to facilitate local language integration in notices.

7.REFERENCES

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